Initial Design Document: RL-Driven Just-in-Time Adaptive Interventions

A Configurable Simulation Framework for Physical Activity

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1 Introduction

Reinforcement learning offers a principled approach to personalising just-in-time adaptive interventions (JITAIs) for physical activity, but training RL agents requires either expensive real-world trials or a realistic simulation environment. While existing simulation environments such as StepCountJITAI [1] and Health Gym [4] provide valuable foundations, they are scoped to specific datasets or research questions. This work supports a cohort study on physical activity and health behaviour change, building a simulation framework to test intervention strategies before real-world deployment. We present `rl-health-interventions`, a configurable simulation framework for just-in-time adaptive interventions (JITAIs): the dataset schema, MDP dynamics, user behaviour models, and agent configurations are specified entirely through YAML config files—no source code changes required. This enables systematic comparison of RL agents on PA intervention tasks using synthetic data in Phase 1, with real-data integration planned for Phase 2.

The initial implementation delivers a rule-based user simulator with four behavioural archetypes, a multi-timescale MDP environment (daily steps + delayed 3-week clinical measures), and Thompson Sampling as the baseline agent [1, 2]. Deep RL agents (DQN, PPO) will be added if time permits. All components share a common ABC + registry interface for extensibility.

Research objectives. We aim to investigate: (1) which state features best predict user response to interventions, (2) whether Thompson Sampling outperforms fixed and rule-based baselines under heterogeneous user archetypes, (3) how short-term wearable signals and long-term health outcomes can be combined in a multi-timescale reward, and (4) whether LLM-simulated users provide realistic evaluation environments for RL policy testing. These questions are exploratory; the framework is a tool for investigation, not a guaranteed outcome.

2 Literature Review

Reinforcement learning has been applied to health interventions through micro-randomized trials (MRTs), most notably HeartSteps V1 and V2, which delivered contextually-tailored activity suggestions and demonstrated the feasibility of RL-in-the-loop policy learning [2, 3]. These trials provide the only available source of real action-to-response mappings for calibrating user behaviour models.

Existing simulation environments for RL-driven health interventions tend to be tied to specific datasets or tasks. StepCountJITAI [1] provides a simulation environment for physical activity JITAIs but is scoped to a single dataset and action set. Health Gym [4] offers synthetic health environments but does not support configurable MDP specifications. Open wearable benchmark datasets such as ExtraSensory [5] and WISDM [6] are valuable for pipeline testing but lack intervention response data.

The gap we address is a config-first simulation framework where the dataset schema, MDP dynamics, user behaviour, and agent configurations are specified entirely through config files, enabling systematic cross-dataset and cross-policy comparisons without bespoke engineering.

A comprehensive literature review with 84 papers is in `docs/sources/`.

3 MDP Formalisation

This section specifies the Markov Decision Process (MDP) for the first iteration of the simulation framework. The MDP is defined as the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

Implementation assumptions. We assume full observability: the agent observes the complete state s_t at each decision point. Episode length and action constraints (e.g., maximum interventions per day, minimum rest periods) are configurable parameters, allowing the framework to adapt to different study designs and intervention protocols.

State. The state represents the current context of a user at decision epoch t. States are constructed from wearable-device signals, user profile information, and intervention history:

$$s_t = \{\mathbf{x}_t^{\text{wearable}}, \mathbf{x}_t^{\text{context}}, \mathbf{x}_t^{\text{history}}, \mathbf{x}^{\text{profile}}\}$$

See Appendix A for the full list of state variables. Variables are normalised to $[0, 1]$ before being passed to the agent. The body measure is only observed every 3 weeks (sparse reward signal).

Actions.

$$\mathcal{A} = \{a_0, a_1, a_2, a_3, a_4, a_5\}$$

Table 1: Action space

Action	Label	Description
a_0	No message	Do not intervene
a_1	Motivational prompt	Generic encouragement
a_2	Walking suggestion	Recommend a short walk
a_3	Goal reminder	Remind of step goal
a_4	Recovery suggestion	Stretch or rest
a_5	Progress feedback	Report goal progress

Exactly one action per decision epoch. The "no message" action is critical for managing notification burden.

Each action carries two scalar penalties in the config schema:

- `reward_penalty` — subtracted from the activity-change reward at the current epoch.
- `burden_penalty` — added to the user's cumulative burden (fatigue model).

The "no message" action has zero on both axes; all other actions have positive values. Penalties are configurable per action, allowing researchers to model different notification intensities (e.g. a walking suggestion may have higher burden cost than a motivational prompt). Whether these two penalties should be merged into a single mechanism is an open question — see Decision Log §6.

Transition. The transition function models the probability over next states:

$$P(s_{t+1} \mid s_t, a_t) : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$$

In Phase 1 (rule-based), transitions follow a behavioural response model:

$$\Delta\text{steps}_{t+1} = f(s_t, a_t, \theta_{\text{user}})$$

where θ_{user} parameterises one of four behavioural archetypes:

- **Goal-driven:** Responds to reminders and feedback. Proportional to goal progress.
- **Social responder:** Responds to motivational prompts. Models social desirability bias.
- **Resistant:** Low response. Fatigue accumulates quickly.
- **Stable maintainer:** Already active. Low marginal gain.

Burden evolves as:

$$\text{burden}_{t+1} = \max(0,\ \text{burden}_t - \delta \cdot \mathbb{1}[a_t = a_0] + \text{burden penalty}_{a_t})$$

where a_0 is the "no message" action, $\delta > 0$ is the configurable burden decay rate, and $\mathbb{1}[\cdot]$ is the indicator function. Burden decays only when no intervention is sent, allowing user recovery.

When burden $> B_{\max}$, response probability decays linearly. This is a simple linear-accumulator model. The closest published precedent is StepCountJITAI's bounded habituation variable [1], which uses a $[0,1]$ increment/decrement formulation. The most appropriate burden model for PA interventions is an open question — see Decision Log §6.

Reward. The reward has an immediate and a delayed component.

Immediate reward:

$$R_t^{\text{immediate}} = \alpha \cdot \Delta\text{activity}_t - \beta \cdot \text{reward\_penalty}_{a_t} + \lambda \cdot \text{goal\_progress}_t$$

where $\Delta\text{activity}_t$ is a configurable activity metric (step count by default; active minutes, METs, or a composite can be substituted via config). Whether steps is the most appropriate primary metric is an open question — see Decision Log §6.

Delayed reward (body measures) arrives every 3 weeks ($t = 21k$, i.e. every 21 days):

$$R_{21k}^{\text{delayed}} = \eta \cdot \text{BM\_improvement}_{21k}$$

Whether this sparse formulation is optimal, or whether a decaying reward (e.g. $\eta \cdot \text{BM\_improvement} \cdot \gamma^{21k-t}$ between body measures) would improve learning, is an open question — see Decision Log §6.

Total reward:

$$R_t = \begin{cases} R_t^{\text{immediate}} + R_t^{\text{delayed}} & t \equiv 0 \pmod{21} \\ R_t^{\text{immediate}} & \text{otherwise} \end{cases}$$

The compound reward (immediate + delayed) is a central research question. Weights $(\alpha, \beta, \lambda, \eta)$ are configurable experiment parameters.

Discount factor. $\gamma \in [0.9, 0.99]$. A high discount factor encourages long-term behaviour change over short-term step increases. Appropriate for health habit formation. The optimal range for PA interventions is an open question — see Decision Log §6.

Decision epoch. Daily decisions. The framework provides a base library of policies; per-user policy refinement operates within these constraints. Daily decisions align with typical intervention cadence, the 3-week body measure delay (21 epochs), and practical deployment constraints. Whether hourly decisions would be more appropriate is an open question — see Decision Log §6.

4 Framework Design

The framework is organised into six modules that form a pipeline from configuration to results. All extension points use an ABC + registry pattern: implement an abstract class, register it with a key, and reference that key in config — no source code changes required to add new datasets, agents, or behaviour models.

A typical experiment starts with a YAML file that selects the dataset, MDP parameters, user profiles, and agent. The framework wires everything from config — adding a new dataset or agent means writing a new config (and optionally a new ABC subclass), not touching framework code. All dataset schemas, MDP parameters (state variables, action space, reward weights, transition models), user archetypes, and agent hyperparameters are configurable via YAML. Adding new component types (e.g., novel transition models, new agents) requires Python implementation but follows a standard interface pattern.

Table 2: Software modules and their dependencies

Module	Role	Depends on
Config	YAML → Pydantic schemas, 3-layer validation	—
Data	Polars lazy reader, FeaturePipeline, StateView	Config
MDP	Environment step/reset, Transition ABC, Reward ABC	Data
UserSim	UserProfile (4 archetypes), ResponseModel ABC	MDP
Agent	Agent ABC, ThompsonSampling (+ optional DQN)	MDP
Experiment	Factory wiring, CLI, results output	All above

The ExperimentFactory reads a composed experiment config, instantiates each component in dependency order, and runs the trial loop. Results are written to CSV with a YAML sidecar for reproducibility. The ABC + registry pattern means swapping any component requires only a new subclass and a config key — no framework modifications. See `docs/code/codebase_plan.md` and `docs/code/code_design.md` for the full specification and `README.md` for a milestone/issue dependency map.

5 Data Availability

This project uses a three-tier data strategy. **Phase 1** combines synthetic data generators (parameterised from published population statistics) with seven immediately-accessible open datasets for pipeline validation: PMData (1,836 rows, Kaggle), HARTH (8.3M rows, HuggingFace), WESAD (60.8M rows, Kaggle), WISDM (1.1M rows, direct download), Fitbit Tracker (940 rows, Kaggle), 4TU Step Goals (117 rows, direct download), and synthetic generation. These datasets provide real wearable data for testing the ingestion pipeline and parameterising synthetic generators before Phase 2 access is granted.

Phase 2 targets three large-scale datasets requiring institutional applications: HeartSteps V1/V2 [2, 3] provide MRT data with intervention response mappings for calibrating the user simulator. The All of Us Fitbit Dataset [7] offers 59,000+ participants with wearable data and EHR linkage (4-8 week application, cloud-only access). The UK Biobank Accelerometer Dataset [8] provides 700,000+ person-days of raw accelerometer data (4-8 week application).

Detailed schema documentation and access procedures are in `docs/sources/data_availability_schema.md` Feasibility studies for Phase 2 datasets are in `docs/sources/data_sources.md` and `docs/sources/additional`

6 Decision Log

Table 3: Config

Question	Decision	Status
What config format?	YAML	Confirmed
What interface?	CLI + config files; web UI = stretch	Confirmed
Which language and package manager?	Python 3.11 + `uv`	Confirmed
Which dependencies?	Minimal: no Gymnasium, no SB3	Confirmed
How to distribute?	Both: repo-based now, package if needed later	Confirmed
Is the config-driven boundary correct?	Current boundary: all parameters configurable via YAML, new component types require Python implementation. Verify this is the right balance between flexibility and implementation effort.	Open

Table 4: Data

Question	Decision	Status
What activity metric for the immediate reward?	Steps (default). Active minutes, METs, or composites are configurable alternatives.	Open
What format for experiment output?	CSV, terminal table, JSON, or summary plots	Open
Which datasets for Phase 2?	HeartSteps first, then All of Us, UK Biobank (see §5)	Confirmed
What data for Phase 1?	Synthetic data + 7 open datasets for pipeline validation (see §5)	Confirmed

Table 5: MDP

Question	Decision	Status
What reward structure?	Multi-timescale: immediate + 3-week delayed	Confirmed
How are components wired?	ABC + registry pattern	Confirmed
What is the decision epoch cadence?	Daily (hourly is a configurable alternative). To be validated empirically.	Open
Is MDP formalisation reasonable?	Awaiting supervisor approval	Open
Which burden model is most appropriate?	Linear accumulator $\max(0, burden_t - \delta \cdot \mathbb{1}[a_t = a_0] + penalty)$ with threshold decay. Configurable decay rate δ. StepCountJITAI uses bounded $[0, 1]$ habituation.	Open
What discount factor range is optimal?	$\gamma \in [0.9, 0.99]$ to be explored empirically	Open
How should action penalties relate?	Currently separate mechanisms: reward_penalty (direct reward subtraction) and burden_penalty (state-accumulator addition), configurable per action and per archetype. Merging is possible if simpler.	Open
Sparse vs decaying delayed reward?	3-week sparse (current). Decaying formulation ($\eta \cdot BM \cdot \gamma^{21k-t}$) may improve credit assignment.	Open

Table 6: Agent

Question	Decision	Status
Which agent first?	Thompson Sampling	Confirmed
What success criteria for agent comparison?	Minimum improvement threshold (e.g., ≥15% reward gain over best baseline) to be determined after initial experiments.	Open
What evaluation methodology?	Baselines (random, fixed, rule-based), metrics (regret, reward, adherence, sustained change), statistical analysis (bootstrap CIs, effect sizes) to be defined before Sprint 3.	Open
Metrics for success?	Output metrics to evaluate agent performance (regret, reward, adherence)	Open

Table 7: User Simulation

Question	Decision	Status
Which user archetypes?	Goal-driven, social, resistant, stable maintainer	Confirmed
Are user archetypes evidence-based?	Current 4 archetypes are a Phase 1 simplification. Validate against real behavioural data in Phase 2 (HeartSteps, All of Us) and refine if needed.	Open

Table 8: Safety & Ethics

Question	Decision	Status
What safety constraints for health interventions?	Hard limits (max intervention frequency, min recovery period, burden threshold) to be implemented in Sprint 4.	Open
How to handle ethics/privacy for Phase 2 data?	IRB approval status, GDPR/HIPAA compliance, data use agreements to be documented before real data access.	Open
What is the treatment policy constraint?	Treatment policies are predetermined by the doctor. The RL agent can only vary within the constraints of that policy, not pick an arbitrary alternative. Clarify how this constraint interacts with the action space and personalisation.	Open

7 Phase 2: Real-Data Validation

Phase 2 shifts from synthetic to real data, calibrating the user simulator and grounding MDP dynamics against observed behavioural responses (see §5). Phase 1 scope is fixed and must be completed before any stretch goals are attempted. Beyond Phase 2, stretch goals include LLM-based user simulation, offline RL (CQL, IQL), and non-stationary environment adaptation. User behaviour clustering will inform the LLM simulation design, enabling persona-based response models grounded in observed behavioural patterns.

Implementation roadmap. A comprehensive audit (June 2026) generated 43 issues across 4 domains: 10 milestone issues (M-01 through M-10), 13 risk issues (including 5 RL-specific risks), 8 documentation debt issues, and 12 stub implementation issues. The roadmap spans 4 sprints (12–16 weeks): Sprint 1 (config schemas, StateView/Environment, synthetic data, CI/CD), Sprint 2 (transition/reward models, user simulation, Thompson Sampling), Sprint 3 (experiment runner, evaluation framework, documentation), and Sprint 4 (safety constraints, ethics documentation, clinical validity). Full audit reports available in `reports/` directory and `AUDIT_MASTER.md`.

References

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A State Variables

Table 9: State variables

Variable	Type	Description
steps_t	$\mathbb{R}_{\geq 0}$	Step count, past 24h
hr_t	$\mathbb{R}_{\geq 0}$	Heart rate (or resting)
sleep_hours_t	$\mathbb{R}_{\geq 0}$	Previous night sleep
sedentary_min_t	$\mathbb{R}_{\geq 0}$	Sedentary minutes today
time_of_day_t	$\{0, \ldots, 23\}$	Current hour
day_of_week_t	$\{0, \ldots, 6\}$	Day of week
goal_progress_t	$[0, 1]$	Fraction of daily goal met
burden_t	$\mathbb{R}_{\geq 0}$	Cumulative burden (decays when no interventions)
a_{t-1}	$\mathcal{A}$	Previous action
response_{t-1}	$\{0, 1\}$	Previous response
body_measure_k	$\mathbb{R}$	Clinical measure (q3wks)
age	$\mathbb{Z}_{>0}$	Static profile
gender	$\{0, 1\}$	Static profile
baseline_activity	$\{0, 1, 2\}$	Low / medium / high